

CURBCOUNT · DESIGN PRECIS

CurbCount design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
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Source	docs/02-concept.md @ 141e630		

REVISION HISTORY

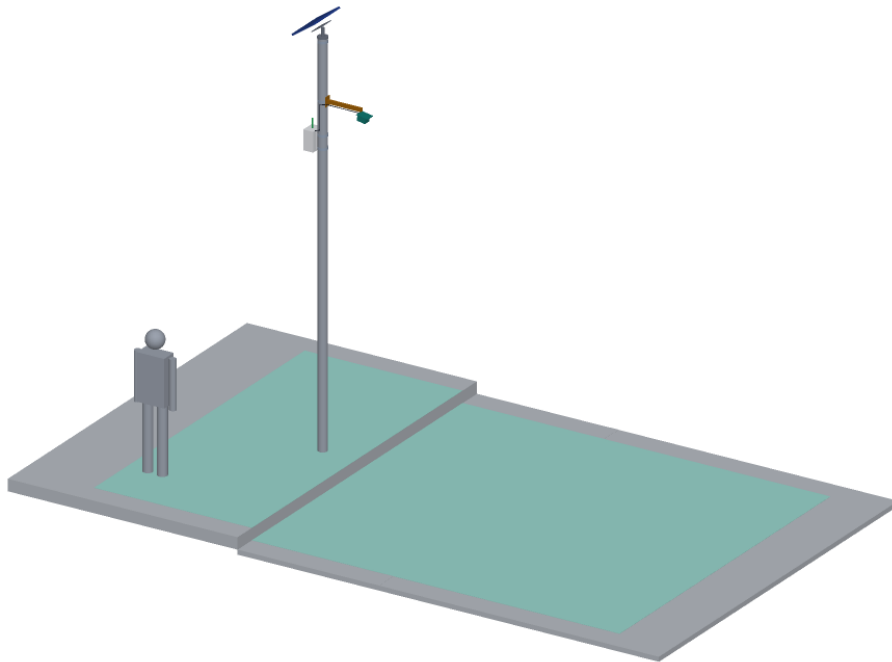
Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CurbCount design precis

CurbCount is a clamp-on street pole counter. A 32 x 24 pixel thermal array looks down from about 4.3 m over the sidewalk, bike lane and nearest traffic lane; an edge processor turns the heat blobs into counts of people, cyclists and vehicles by direction; and a FieldNode core sends only 15-minute counts over LoRaWAN. Each pixel covers about 0.3 m of ground, so the sensor cannot show a face or a number plate. First-order numbers suggest it can count continuously for about 4 days without sun, but the parts cost of about \$271 is well over the \$150 budget.

CurbCount

CONCEPT, NOT FOR FABRICATION



Grey: Street pole, 5 m, Sidewalk and road, Sensing footprint (estimate), Person, 1.75 m for scale

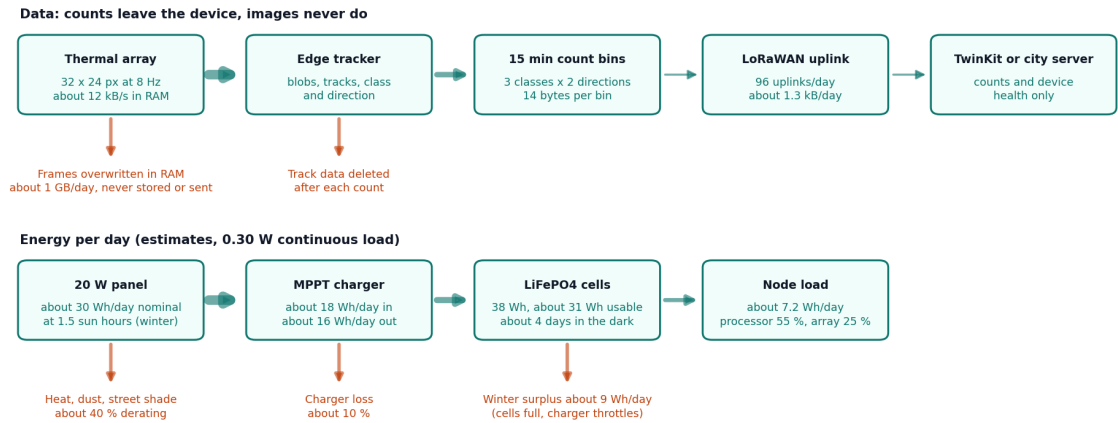
■ How it works

1. **Sense.** The thermal array reads 32 x 24 temperatures eight times a second. People, cyclists and engines stand out from the pavement as warm (or, on hot pavement, cool) blobs.
2. **Track.** The edge processor keeps a slowly updated background, subtracts it, finds blobs, and follows each blob across frames. Frames live only in RAM and are overwritten about every 125 ms.
3. **Classify.** Each finished track is classed as pedestrian, cyclist (including e-scooters and other micromobility) or motor vehicle from its size, speed and position (sidewalk, bike lane or traffic lane), and given a direction along the street.

4. **Count.** Tracks increment six counters (three classes by two directions); the track data is then deleted. Every 15 minutes the counters close as one 14-byte record.
5. **Send.** The FieldNode core sends each record over LoRaWAN to the lab's TwinKit gateway, a city server or The Things Network, with a daily health message (battery, temperature, frame errors).

CurbCount: data and energy flow

CONCEPT, NOT FOR FABRICATION. Values are estimates to be checked at TRL 3.



■ **Main components**

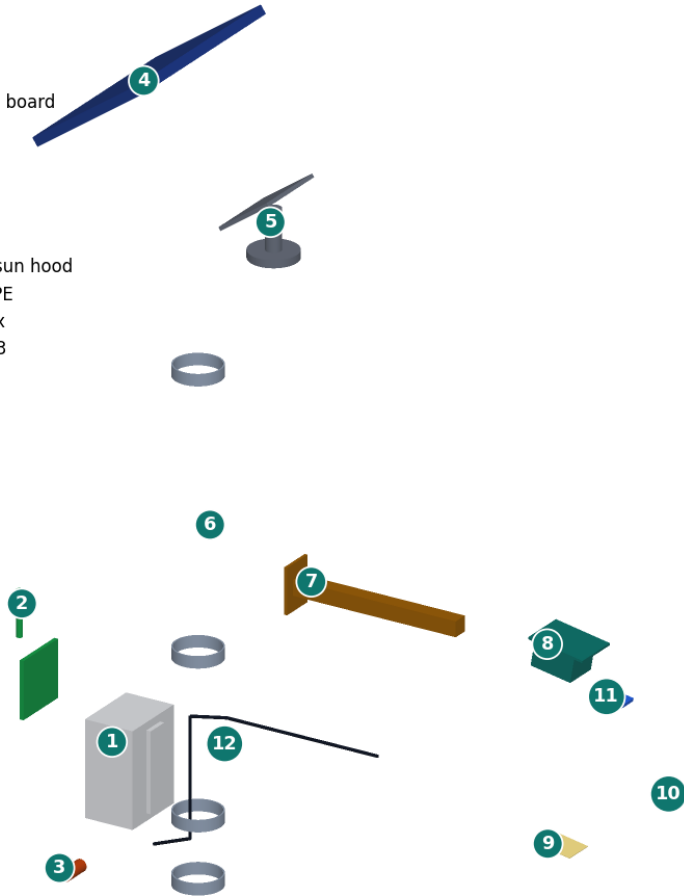
Table 1. Main components. Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

#	Component	Proposed choice	Notes
1	FieldNode enclosure	IP65 polycarbonate box, about 180 x 130 x 250 mm, vent and gland	From FieldNode, unchanged
2	FieldNode power and radio board	MPPT charger with cold-charge lockout, STM32WL-class LoRaWAN module, switched sensor rail	From FieldNode, unchanged
3	LiFePO4 cells	Two 3.2 V 6 Ah cells (FieldNode's one plus one added)	Added cell proposed, awaiting Amish
4	Solar panel	20 W, about 540 x 350 mm, in place of FieldNode's 6 W	Proposed, awaiting Amish
5	Panel tilt bracket	Pole-top cap and hinge, about 35° tilt	Adapted from FieldNode
6	Band clamps	Four stainless band clamps with rubber liners	No drilling
7	Sensor arm	40 x 40 mm aluminum tube, about 450 mm reach toward the road	Keeps the head clear of the pole shadow
8	Sensor head housing	3D-printed ASA, about 110 x 90 x 70 mm, with a sun hood	Tilted 10° toward the road
9	Window	0.5 mm HDPE film, which passes long-wave infrared	Far cheaper than germanium; transmission to be checked
10	Thermal array	MLX90640 class, 32 x 24 px, 110 x 75° lens	Proposed, awaiting Amish (see radar option)
11	Edge processor	ESP32-S3 class module, radio disabled in firmware	Proposed, awaiting Amish
12	Sensor cable	M12 sealed cable to a FieldNode sensor port	FieldNode standard pinout

CurbCount: exploded view

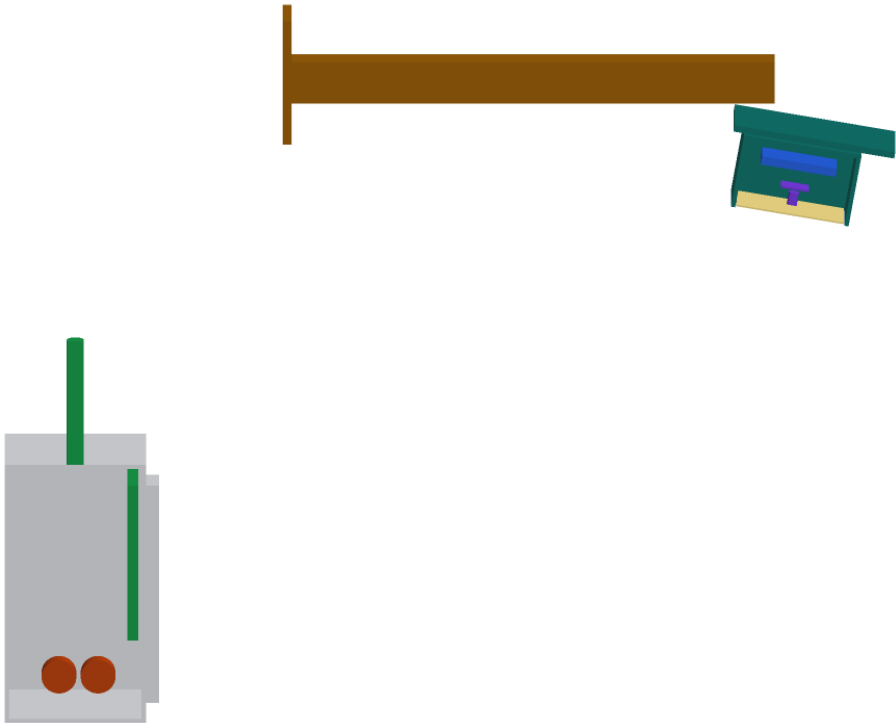
CONCEPT, NOT FOR FABRICATION

- 1 FieldNode enclosure, IP65
- 2 FieldNode power and radio board
- 3 LiFePO4 cells, 2 x 6 Ah
- 4 Solar panel, 20 W
- 5 Panel tilt bracket
- 6 Stainless band clamps (4)
- 7 Sensor arm, aluminium
- 8 Sensor head housing and sun hood
- 9 LWIR window, 0.5 mm HDPE
- 10 Thermal array, 32 x 24 px
- 11 Edge processor, ESP32-S3
- 12 Sensor cable, M12



CurbCount: cutaway

CONCEPT, NOT FOR FABRICATION



■ **First-order numbers**

All values are estimates for concept review and will be checked at TRL 3.

Table 2. First-order numbers.

Quantity	Estimate	Basis and assumptions	Requirement
Footprint across the street	about 6.9 m, from 2.1 m behind the curb to 4.8 m into the road	75° field across, window 4.3 m above the road, tilted 10° toward the road	R6 met for sidewalks to 2 m; not met for 3 m
Footprint along the street	about 12 m	110° field along, at 4.3 m	
Ground size of one pixel	about 0.3 to 0.4 m	6.9 m / 24 px across; 12.3 m / 32 px along, at nadir	R4 met (no face or plate can be resolved)
Frames per crossing	about 14 for a cyclist at 25 km/h; about 7 for a car at 50 km/h	12 m footprint, 8 Hz frame rate	R1 plausible
Raw frame data	about 12 kB/s, about 1 GB/day, held only in RAM	768 px x 2 bytes x 8 Hz	R4
Uplink data	14 bytes per 15 min, 96 uplinks/day, about 1.3 kB/day	3 classes x 2 directions x 2 bytes, plus status	R5 met
Radio airtime	about 0.2 s per uplink, about 0.8 s/h	SF9 at 125 kHz, about 27 bytes with LoRaWAN overhead; EU868 sub-band limit 1 % (36 s/h)	R15 met
Thermal array power	about 0.07 W	About 20 mA at 3.3 V (typical class value, to be confirmed)	
Edge processor power	about 0.17 W	About 50 mA at 3.3 V with its own radio off	
Conversion and base load	about 0.06 W	Regulator losses about 15 %; FieldNode base about 0.02 W	
Total continuous load	about 0.30 W, about 7.2 Wh/day	Counting 24 h a day	
Share of FieldNode sensor budget	about 2.4 times	FieldNode README allows about 115 mW average for sensors	Standard FieldNode not sufficient
Battery autonomy	about 4.3 days with two cells; about 2.1 days with one	2 x 19.2 Wh, 80 % usable	R8 met with two cells only
Winter solar harvest	about 16 Wh/day into the cells	20 W x 1.5 peak sun hours x 0.6 derating x 0.9 charger	R9 met (about 2.2 times the load)
Same with FieldNode's 6 W panel	about 4.9 Wh/day	Same assumptions	R9 not met
Mass on the pole	about 4.2 kg	FieldNode about 1.7 kg, larger panel +1.0 kg, cell +0.2 kg, head 0.35 kg, arm 0.6 kg, clamps 0.3 kg	R11 met

Quantity	Estimate	Basis and assumptions	Requirement
Wind load on panel	about 170 N	0.19 m ² , 35 m/s gust, force coefficient 1.2	R12, to check bracket
Parts cost	about \$271	Indicative prices, see <code>bom/bom.csv</code>	R13 not met

The footprint and pixel size come from simple geometry over flat ground. Real streets have parked cars, trees and awnings that hide parts of the footprint, and people walking side by side will merge into one blob, so counts will run low in dense flows until the tracker handles merged blobs.

■ Key design choices

- **Thermal array rather than a camera.** At 32 x 24 pixels the sensor cannot capture an identifiable image at any mounting height used here, so privacy does not depend on trusting the firmware. Proposed, awaiting Amish.
- **Thermal array rather than 60 GHz radar.** A thermal array is cheaper, simpler and lower power, and a radar is better in heat, fog and darkness and at separating people from bicycles by motion. Proposed: thermal for the first build, radar kept as a variant, awaiting Amish.
- **Frames never leave RAM.** The firmware has no code path that writes or sends a frame. A calibration mode that shows frames on a laptop through a physical jumper is proposed for setup only, awaiting Amish.
- **FieldNode core with a larger panel and a second cell.** The counter draws about 2.4 times FieldNode's sensor budget. Proposed, awaiting Amish; the change should be raised with the FieldNode project as a "high-load" variant, not made to FieldNode here.
- **Separate processor in the head.** Keeping tracking next to the sensor means only counts cross the M12 cable. Running the tracker on FieldNode's own STM32WL instead would save about \$12 and most of the 0.17 W, but may be too slow for tracking at 8 Hz. Proposed: ESP32-S3 first, STM32WL-only as a TRL 3 study, awaiting Amish.
- **15-minute bins.** This matches common count practice and keeps radio airtime tiny. Proposed, awaiting Amish.

■ Safety

Safety: Installing on a street pole is work at height next to traffic. Install only with the pole owner's permission, by trained crews, with fall protection and traffic management as local rules require. Keep clear of overhead power lines and street-light wiring, and never open a pole's electrical hatch.

Safety: The node contains two LiFePO4 cells. Fuse the pack, charge only within the cell maker's temperature limits (FieldNode's cold-charge lockout applies), and do not install a node with a swollen or damaged cell.

Safety: A falling part from 4 to 5 m can injure people below. Use a secondary safety lanyard on the panel and the sensor head, and check clamps after the first storm.

Privacy. No images, audio recordings or personal identifiers leave the device; only aggregate counts are stored or sent. Check local data protection law before any deployment, and publish a notice on the pole saying what is counted and where the design is documented.

■ Open questions for TRL 3

- Measure HDPE window transmission at 8 to 14 μm and the resulting loss of contrast.

- Test detection when pavement and air are near body temperature (about 30 to 35 °C); decide whether the radar variant is needed for hot climates.
- Define the tracker for merged blobs (groups, parents with strollers) and the target accuracy per class.
- Confirm the MLX90640 and ESP32-S3 currents at 8 Hz and whether the STM32WL alone can run the tracker.
- Check street shading of the panel on real poles, which may cut harvest more than the 40 % derating assumed.
- Choose the first co-design partner and street.

Concept media: [blueprint sheet](#), [interactive 3D model](#).